DEEP GENERATIVE MODELING FOR LABEL NOISE DETECTION

Cheng Lun Tai
Department of Electrical and Computer Engineering
University of California, San Diego
c8tai@ucsd.edu

Abstract

Noisy labels are common in large datasets and can harm the performance of machine learning models. In this work, we propose a deep generative model that explicitly models the process of label corruption and detects potentially incorrect labels through posterior inference. Our model introduces latent variables for the clean label and image generation, while the observed label is produced through a corruption matrix. Using variational inference, we estimate the posterior distribution of the clean label and compute a noise score that measures the disagreement between the image and its observed label. Experiments on CIFAR-10 with both symmetric and asymmetric label noise show that the proposed method improves noise detection compared to several baseline variants.

1 Introduction

Large datasets are essential for modern machine learning systems, especially in computer vision and natural language processing. However, real-world datasets often contain noisy labels due to crowdsourcing, automated collection, or weak supervision. Incorrect labels can degrade model performance and introduce bias, particularly when standard supervised learning assumes labels are correct [2, 3, 4].

A common approach is to modify the loss function or estimate a noise transition matrix. While effective, these methods typically treat label noise as a nuisance rather than modeling the data generation process.

In preliminary experiments on CIFAR-10, we observed that standard classifiers tend to overfit noisy labels. As training progresses, the model may confidently predict incorrect labels even when the visual content clearly suggests otherwise.

This motivates a generative formulation. We treat the clean label as a latent variable that generates the image, while the observed label is produced through a corruption process. This allows the model to evaluate whether an observed label is consistent with the image.

We use variational inference with neural networks to approximate the posterior distribution over latent variables. The posterior probability of the observed label provides a natural measure of label consistency, which we use to define a noise score.

Experiments on CIFAR-10 show improved noise detection under both symmetric and asymmetric label noise. In particular, removing noisy-label conditioning from the posterior leads to more robust inference and clearer separation between clean and corrupted samples.

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2 Background

2.1 Learning with Noisy Labels

Learning with noisy labels has been widely studied in machine learning. Traditional approaches often focus on modifying the loss function to reduce the impact of incorrect labels. For example, loss correction methods attempt to adjust the training objective using an estimated noise transition matrix. [2, 3] Another line of work estimates the corruption matrix that describes how true labels are transformed into observed noisy labels.

While these methods can improve classification robustness, they typically rely on discriminative models that directly predict labels from inputs. Such models may still overfit noisy labels, especially when the noise level is high.

2.2 Deep Generative Models

Deep generative models provide a probabilistic way to model complex data. Latent variable models such as variational autoencoders (VAEs) introduce variables that capture hidden structure in the data. [1] Training is typically performed using variational inference by maximizing the evidence lower bound (ELBO).

By modeling how data is generated, generative models can reason about whether observations are consistent with latent variables. This property makes them particularly suitable for tasks such as anomaly detection and data quality assessment.

3 Related Work

Several methods have been proposed to address the problem of label noise. Early approaches focus on estimating a noise transition matrix that models how clean labels are corrupted into noisy labels. These methods often rely on additional assumptions or clean validation sets to estimate the corruption process.

Another line of research attempts to identify and filter out noisy samples using prediction confidence or agreement among multiple models. Although these methods can be effective in practice, they do not explicitly model the relationship between the true label, the observed label, and the input data.

Recent work has explored the use of probabilistic models to jointly model data and label noise. By introducing latent variables representing clean labels, these approaches allow the model to infer whether an observed label is consistent with the data. Our work follows this direction by using a deep generative model combined with variational inference to estimate the posterior distribution of the true label.

4 Method

Unlike a purely discriminative classifier, our model defines a generative process over images, clean labels, and noisy labels. The key idea is to treat the true label as a latent variable and generate the observed label through a corruption process. Using variational inference, the model can estimate whether the observed label is consistent with the image.

4.1 Generative Model

We assume that each data point is generated through the following probabilistic process:

$$y \sim p(y) \tag{1}$$
$$z \sim p(z) \tag{2}$$
$$x \sim p_\theta(x \mid z, y) \tag{3}$$
$$\tilde{y} \sim p(\tilde{y} \mid y) \tag{4}$$

Here, x denotes the observed image, y is the latent clean label, z is a continuous latent variable capturing additional variation in the image, and $\tilde{y}$ is the observed (potentially corrupted) label. The corruption process $p(\tilde{y} \mid y)$ is modeled using a label transition matrix C, where C_{ij} represents the probability that a true label i is observed as label j.

We assume a uniform prior over labels:

$$p(y) = \frac{1}{K}, \tag{5}$$

where K is the number of classes. The latent variable z follows a standard Gaussian prior:

$$p(z) = \mathcal{N}(0, I). \tag{6}$$

The conditional distribution $p_\theta(x \mid z, y)$ is parameterized by a neural decoder network. In our implementation, we concatenate the latent variable z with a one-hot representation of the label y and feed the result into a deconvolutional network that generates image logits.

4.2 Graphical Model

The structure of the proposed model can be illustrated using a graphical model. The true label y and latent variable z jointly generate the image x, while the observed label $\tilde{y}$ is produced from the true label through a corruption process.

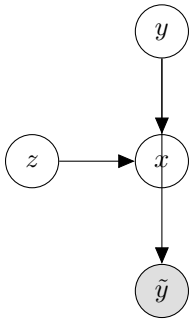

Figure 1: Graphical model of the proposed generative process. The latent clean label y and latent variable z generate the image x, while the observed label $\tilde{y}$ is produced through a corruption process.

This structure explicitly separates the clean label from the observed noisy label, allowing the model to reason about whether the observed label is consistent with the image.

4.3 Variational Inference

Given an observed image and noisy label $(x, \tilde{y})$, we aim to infer the posterior distribution over the latent variables:

$$p(y, z \mid x, \tilde{y}). \tag{7}$$

However, this posterior is intractable due to the nonlinear neural networks in the generative model. We therefore use variational inference to approximate the posterior using neural networks.

Specifically, we introduce the following variational distributions:

$$q_\phi(y \mid x), \tag{8}$$
$$q_\phi(z \mid x, y). \tag{9}$$

The distribution $q_\phi(y \mid x)$ is parameterized by a convolutional neural network that outputs a categorical distribution over the possible labels. This network corresponds to the `QyNet` in our implementation.

Conditioned on the image and label, the distribution $q_\phi(z \mid x, y)$ is parameterized by another neural network that outputs the mean and variance of a Gaussian distribution. This network corresponds to `QzNet`, which models the posterior over the continuous latent variable z.

We made a critical design choice here. By intentionally decoupling the posterior y from the noisy label $\tilde{y}$, we prevent the inference network from "inheriting" the very corruption we aim to detect. The rationale is simple: the image should speak for itself. Conditioning on the observed label may cause the inference network to over-trust corrupted labels. By using $q_\phi(y \mid x)$ instead of $q_\phi(y \mid x, \tilde{y})$, the model relies primarily on visual evidence when inferring the latent clean label.

4.4 Noise Detection Score

The posterior distribution over the clean label provides a natural way to detect label noise. If the observed label $\tilde{y}$ is inconsistent with the image, the posterior probability assigned to that label will be small.

We therefore define a noise score as

$$s(x, \tilde{y}) = 1 - q_\phi(y = \tilde{y} \mid x). \tag{10}$$

A higher score indicates that the observed label is less consistent with the image under the inferred posterior distribution. This score can be used to rank samples according to the likelihood that their labels are corrupted.

5 Training Objective

Given an observed image-label pair $(x, \tilde{y})$, our goal is to maximize the marginal log-likelihood

$$\log p(x, \tilde{y}). \tag{11}$$

However, this quantity is intractable because it requires marginalizing over the latent clean label y and the continuous latent variable z:

$$p(x, \tilde{y}) = \sum_y \int p(x, \tilde{y}, y, z)\, dz. \tag{12}$$

To make learning tractable, we use variational inference and optimize the evidence lower bound (ELBO). Starting from

$$\log p(x, \tilde{y}) = \log \mathbb{E}_{q_\phi(y,z|x,\tilde{y})} \left[\frac{p(x, \tilde{y}, y, z)}{q_\phi(y, z \mid x, \tilde{y})} \right], \tag{13}$$

Jensen's inequality gives

$$\log p(x, \tilde{y}) \geq \mathbb{E}_{q_\phi(y,z|x,\tilde{y})} \left[\log p(x, \tilde{y}, y, z) - \log q_\phi(y, z \mid x, \tilde{y}) \right]. \tag{14}$$

Using the factorization of our generative model,

$$p(x, \tilde{y}, y, z) = p(y)\, p(z)\, p_\theta(x \mid y, z)\, p(\tilde{y} \mid y), \tag{15}$$

and the variational approximation

$$q_\phi(y, z \mid x, \tilde{y}) \approx q_\phi(y \mid x)\, q_\phi(z \mid x, y), \tag{16}$$

the ELBO can be written as

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi(y|x)} \Big[& \mathbb{E}_{q_\phi(z|x,y)} [\log p_\theta(x \mid y, z)] + \log p(\tilde{y} \mid y) + \log p(y) \\ & - \mathrm{KL}\big(q_\phi(z \mid x, y) \,\|\, p(z)\big) \Big] + H\big(q_\phi(y \mid x)\big), \end{aligned} \tag{17}$$

where $H(q_\phi(y \mid x))$ denotes the entropy of the variational posterior over the latent clean label.

Equation (17) has several intuitive components. The reconstruction term

$$\mathbb{E}_{q_\phi(y|x)} \mathbb{E}_{q_\phi(z|x,y)} [\log p_\theta(x \mid y, z)] \tag{18}$$

encourages the decoder to generate images that are consistent with the inferred clean label and latent variable. The corruption term

$$\mathbb{E}_{q_\phi(y|x)}[\log p(\tilde{y} \mid y)] \tag{19}$$

encodes how likely the observed noisy label is under the assumed corruption process. The KL term regularizes the continuous latent variable z toward the prior $p(z)$, while the entropy term prevents the posterior over labels from collapsing too early.

In practice, we use a β-weighted KL objective:

$$\begin{aligned}\mathcal{L}_\beta = \mathbb{E}_{q_\phi(y|x)}\Big[&\mathbb{E}_{q_\phi(z|x,y)}[\log p_\theta(x \mid y, z)] + \log p(\tilde{y} \mid y) + \log p(y) \\ &- \beta\,\mathrm{KL}\big(q_\phi(z \mid x, y) \,\|\, p(z)\big)\Big] + H\big(q_\phi(y \mid x)\big),\end{aligned} \tag{20}$$

where β controls the strength of latent regularization. In our final model, we use $\beta = 0.3$, which empirically improves training stability and yields better noise detection performance.

Since CIFAR-10 contains only $K = 10$ classes, we compute the expectation over $q_\phi(y \mid x)$ by explicitly enumerating all possible labels. For each class k, we infer a Gaussian posterior $q_\phi(z \mid x, y = k)$, sample z, evaluate the decoder likelihood $p_\theta(x \mid z, y = k)$, and weight the result by the posterior probability $q_\phi(y = k \mid x)$. This implementation closely matches the variational objective in Eq. (20).

Finally, the model is trained by minimizing the negative ELBO:

$$\mathcal{J} = -\mathcal{L}_\beta. \tag{21}$$

6 Experiments

6.1 Experimental Setup

We evaluate the proposed model on the CIFAR-10 dataset, which contains 60,000 images across 10 classes.[5] Each image has resolution 32×32.

To study robustness under label corruption, we simulate two types of noisy labels.

Symmetric noise. With a fixed probability, the true label is replaced by a randomly selected incorrect class.

Asymmetric noise. Labels are flipped to semantically similar classes (e.g., cat $\rightarrow$ dog or truck $\rightarrow$ automobile), producing more realistic label noise.

In the symmetric setting, the noise rate is approximately 20% for both training and test sets. In the asymmetric setting, because only a subset of classes is corrupted, the effective noise rate is approximately 10%.

We compare three model variants:

- **Original**: posterior $q(y \mid x, \tilde{y})$ with a learned corruption matrix.
- **Fixed-C**: posterior $q(y \mid x, \tilde{y})$ with a fixed corruption matrix.
- **Final**: posterior $q(y \mid x)$ with a fixed corruption matrix and $\beta = 0.3$.

In the Fixed-C and Final variants, the corruption matrix is pre-defined based on the known noise rate. For symmetric noise, we assume a uniform corruption probability across incorrect classes. For asymmetric noise, the matrix follows the predefined class transition rules used to generate the dataset.

This setup allows us to isolate the effect of posterior modeling without introducing additional uncertainty from estimating the corruption matrix.

Noise detection performance is evaluated using ROC-AUC and PR-AUC.

6.2 Quantitative Results

We also perform an ablation study to understand the contribution of different design choices in the model. Specifically, we compare three variants that differ in how the corruption matrix is modeled and whether the posterior distribution conditions on the noisy label.

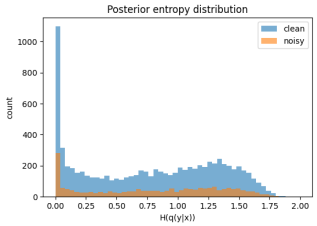

Figure 2: Posterior entropy distribution for clean and noisy samples under symmetric noise. Clean samples typically produce lower entropy values, indicating confident posterior predictions, while noisy samples often exhibit higher entropy due to disagreement between the image content and the observed label.

- **Original**: posterior $q(y \mid x, \tilde{y})$ with a learned corruption matrix.
- **Fixed-C**: posterior $q(y \mid x, \tilde{y})$ with a fixed corruption matrix.
- **Final**: posterior $q(y \mid x)$ with a fixed corruption matrix and $\beta = 0.3$.

Table 1 summarizes the noise detection performance.

Variant	ROC-AUC	PR-AUC
Original	0.612	0.254
Fixed-C	0.638	0.276
Final (Symmetric Noise)	0.720	0.335
Final (Asymmetric Noise)	0.718	0.180

Table 1: Noise detection performance of different model variants.

The ablation study highlights the impact of modeling choices on noise detection performance. Moving from the Original model to Fixed-C improves performance by stabilizing the corruption process. Further improvement is obtained by removing noisy-label conditioning in the posterior, which allows the model to rely primarily on visual evidence when inferring the clean label.

The final model achieves the best performance, improving ROC-AUC from 0.612 to 0.720 under symmetric noise. Overall, the results indicate that removing noisy-label conditioning from the posterior improves robustness. Using a fixed corruption matrix further stabilizes the inference process under label corruption.

6.3 Posterior Entropy Analysis

To better understand the behavior of the model, we analyze the entropy of the posterior label distribution.

As shown in Figure 2, clean samples tend to have lower entropy because the inferred posterior distribution concentrates on a single class. In contrast, noisy samples often produce higher entropy since the image content conflicts with the observed label.

For the asymmetric noise setting, the entropy distribution is shown in Figure 3. The average entropy values are:

- Clean samples: 0.194
- Noisy samples: 0.243

6.4 Noise Score Distribution

We define a noise score

$$s(x, \tilde{y}) = 1 - q_\phi(y = \tilde{y} \mid x), \tag{22}$$

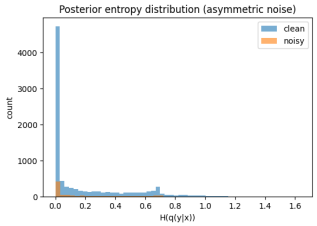

Figure 3: Posterior entropy distribution under asymmetric label noise.

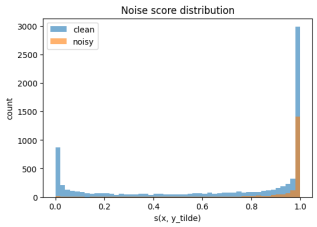

Figure 4: Noise score distribution for clean and noisy samples under symmetric noise. Noisy samples tend to produce higher scores, while clean samples concentrate near zero.

which measures the disagreement between the observed label and the posterior distribution inferred from the image.

As shown in Figure 4, noisy samples generally obtain larger scores, while clean samples remain close to zero. This separation allows the model to rank samples according to their likelihood of label corruption.

For asymmetric noise, the distribution of noise scores is shown in Figure 5.

6.5 Qualitative Examples

To further illustrate how the model identifies corrupted labels, we visualize examples from the asymmetric noise setting.

As shown in Figure 6, the model assigns high noise scores to samples whose visual content clearly contradicts the observed label. In contrast, clean samples typically receive low noise scores when the posterior strongly agrees with the label.

These examples demonstrate that the proposed generative model can distinguish between consistent and inconsistent image-label pairs.

6.6 Discussion

The model still struggles with ambiguity. For visually similar classes (e.g., bird vs. airplane), the posterior may fail to clearly distinguish clean and corrupted labels. This is an expected but important limitation when visual features overlap.

Another limitation is that the generative model is relatively simple compared to modern high-capacity image models. As a result, the reconstruction term may not capture fine-grained details, reducing the sensitivity of the noise detection score in difficult cases.

Despite these limitations, the final model achieves clearer separation between clean and noisy samples in both entropy and noise score distributions. This indicates that the inferred posterior effectively captures inconsistencies between images and observed labels.

Overall, combining latent variable modeling, variational inference, and an explicit corruption process provides a practical approach for detecting label noise.

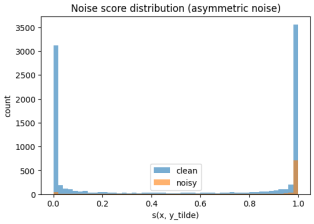

Figure 5: Noise score distribution under asymmetric noise.

(a) Samples with the highest noise scores under asymmetric label corruption.

(b) Examples of clean samples with low noise scores.

Figure 6: Qualitative examples from the asymmetric noise setting. Top: samples with high noise scores, where the image content disagrees with the observed label. Bottom: clean samples with low noise scores, where the posterior strongly agrees with the observed label.

7 Conclusion

In this work, we propose a deep generative approach for detecting noisy labels in images. By modeling both the data generation and label corruption processes, the model can infer the posterior distribution over clean labels.

Experiments on CIFAR-10 show improved noise detection under both symmetric and asymmetric settings. In particular, removing noisy-label conditioning leads to more robust inference and better detection performance.

Future work includes using more expressive generative models, improving corruption modeling, and extending to larger and more complex datasets.

References

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